



OBJECT-ORIENTED DESIGN IN SMART HEALTHCARE SYSTEMS

CSA1104- OBJECT ORIENTED ANALYSIS AND DESIGN

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Introduction



- A Smart Healthcare System is a software application that manages patient and hospital information efficiently.
- It handles appointments, medical records, billing, and doctor details.
- The system uses Object-Oriented Design (OOD) to organize data and functions.
- OOD improves security, code reusability, scalability, and maintainability.
- It helps develop reliable and efficient healthcare management systems.

Class And Object

Class	Object
A class is a blueprint or template used to create objects.	An object is an instance of a class with actual values.
Defines attributes (data) and methods (functions).	Contains real data and performs the defined methods.
It is a logical entity and does not occupy memory until objects are created.	It is a physical entity that occupies memory during execution.
One class can create multiple objects.	Each object has its own unique state and behavior.
Example: Patient, Doctor,	Example: John (Patient), Dr. Smith

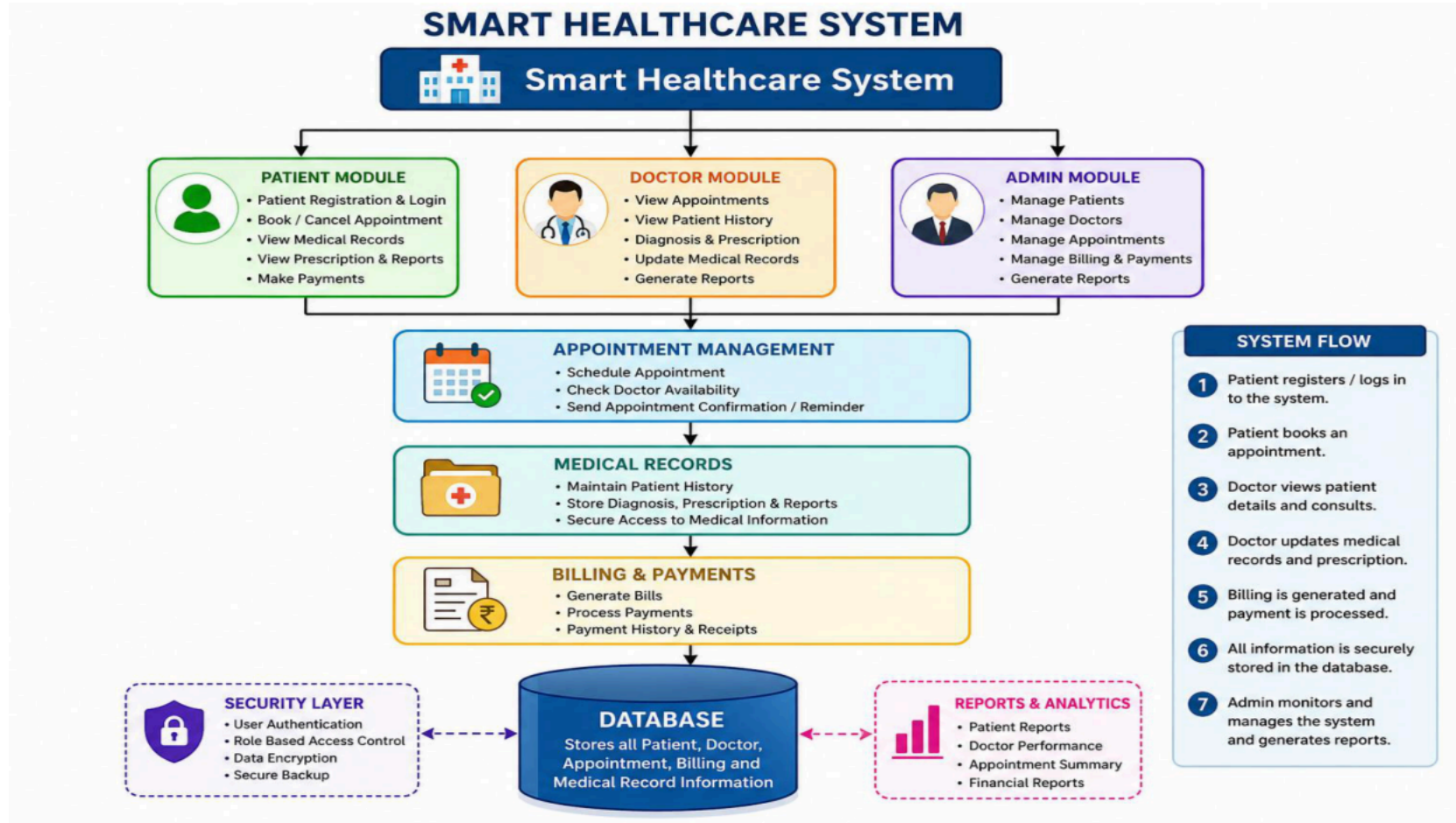
Encapsulati on

Encapsulation	Description
Definition	Combines data and methods into a single unit (class).
Purpose	Protects sensitive data from unauthorized access.
Access Control	Uses private data members and public methods.
Benefit	Improves security, data integrity, and maintainability.
Example	Patient medical records are accessed only through authorized methods.

Inheritance And Polymorphism

Inheritance	Polymorphism
Allows one class to inherit properties and methods from another class.	Allows the same method to perform different actions for different objects.
Promotes code reusability .	Promotes flexibility in programming.
Reduces code duplication.	Improves extensibility and maintainability.
Uses a parent-child relationship between classes.	Uses method overriding or overloading.
Example: Person → Patient, Doctor, Nurse	Example: login() → Patient, Doctor, and Admin perform

System Design / Architecture





Advantages



- **Improved Security** – Protects sensitive patient information using encapsulation.
- **Code Reusability** – Inheritance allows existing code to be reused, reducing duplication
- **Scalability** – New features and modules can be added easily as the system grows.
- **Easy Maintenance** – Modular design makes debugging, updating, and modifying the system simpler
- **Flexibility** – Polymorphism enables different objects to perform tasks in different ways



Conclusion



Object-Oriented Design (OOD) plays a vital role in developing Smart Healthcare Systems by providing a structured, secure, and efficient approach to software development. Concepts such as **class, object, encapsulation, inheritance, and polymorphism** help improve code reusability, data security, scalability, and maintainability. By applying these OOP principles, healthcare applications can efficiently manage patients, doctors, appointments, and medical records while ensuring reliable and high-quality healthcare services.

THANK YOU